

Collaborative Motion Modes in Serious Game Using Virtual Co-embodiment: A Pilot Study on Usability and User Perception

Xiaoyi Xue*

School of Cultural
Technology, Xi'an Jiaotong-
Liverpool University, China

Yangyang He*

School of Cultural
Technology, Xi'an Jiaotong-
Liverpool University, China

Jingjing Zhang

School of Engineering,
University of Liverpool,
United Kingdom

Xiongju Sun†

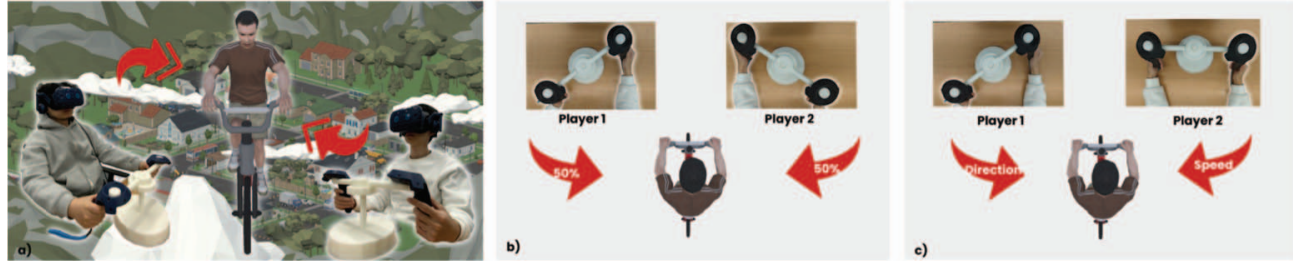
School of Cultural
Technology, Xi'an Jiaotong-
Liverpool University, China

Figure 1: a) A participant and an experimenter jointly control a virtual avatar to accomplish game tasks. b) **weighted-average-based virtual co-embodiment**: The avatar's motion is determined by calculating a weighted average of inputs from two users (e.g., each of the two participants holding a 50% share of influence over both direction and speed). c) **body-part-segmented virtual co-embodiment**: two users have distinct control over specific parts of the avatar's body (e.g., one controls the direction while the other controls the speed).

ABSTRACT

With the increasing attention to collaboration in Serious Games, particularly in immersive virtual environments, a novel approach of virtual co-embodiment (e.g., one avatar controlled by multiple users) in recent studies has the potential to contribute to the research on collaborative motion in multiplayer games. This pilot study investigates the usability and the agency of collaborative motion modes in a virtual cycling game under the virtual co-embodiment system. Results showed that these new collaborative modes reported a higher perceived usability. Besides, based on quantitative and qualitative findings, co-embodiment modes in this virtual serious game might evoke an enhanced users' agency.

Keywords: Virtual Co-embodiment · Serious Game · Usability · Sense of Agency · Collaboration

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Relying on the unique reward mechanisms and highly immersive qualities, Virtual Reality Serious Games (VR SGs) have garnered increasing attention across diverse research domains (e.g., digital VR games for rehabilitation and skill learning) [1]. Incorporating SGs into virtual environments (VEs) has enhanced user engagement and their flow state, particularly when employed within collaborative motion modes in multiplayer gaming settings, further amplifying these effects [2]. As multiplayer collaborative motion modes in VEs become more prevalent, it becomes more

imperative to explore new collaborative motion modes that transcend traditional frameworks. Traditional collaborative motion modes involve multiple individuals independently controlling virtual avatars coexisting in VEs, striving to achieve same or diverse objectives. Recent studies on collaborative training have applied the concept of virtual co-embodiment, defined as a user and another entity jointly controlling a shared virtual avatar [3]. This concept encompasses two configuration methods, namely Weighted-average-based (e.g., a co-embodiment system wherein the control of a virtual avatar is determined by the weighted average of movements from multiple entities) and Body-part-segmented (e.g., a co-embodiment system in which two users concurrently manipulate distinct body parts) virtual co-embodiment. It provides a novel approach to multiplayer collaboration in VEs [4]. However, there is a lack of research applying virtual co-embodiment to the study of VR SGs, particularly in exploring and comparing user experiences within the two collaborative modes in virtual co-embodiment.

This pilot study aims to explore the effects of collaborative motion modes on perceived usability and the agency under the virtual co-embodiment system in a cycling serious game by collecting both quantitative and qualitative evidence. The results of the study can contribute to acquiring higher usability of the new modes, offering insights for the design of virtual co-embodiment-based SGs.

2 USER STUDY

The virtual co-embodiment system used two HTC Vive head-mounted displays (HMDs) with four VR controllers. A participant and an experimenter connect to the same network and inhabit a shared virtual avatar. The system was developed in Unity (version 2021.3.11f1c1) using the Netcode multiplayer networking plugin.

The study was designed for participants to simulate cycling by a virtual body in VR with other people who control the same virtual avatar. The collaborative act of two individuals sharing a bicycle also represents a traditional form of cooperative behavior in the physical world. In this study, the VR controllers that fixed to the

* Joint first author & equal contribution

† Corresponding author: xiongju.sun@gmail.com

simulated bicycle's handlebars could collect steering and speed data of virtual bicycle. When the user rotated the controller, the virtual bicycle responded correspondingly, generating steering movements to simulate the cycling experience. Simultaneously, users utilized the trigger buttons on the controllers to regulate the speed of the bicycle movement. Data processing varies based on the three modes: one being the No collaboration - **Alone mode (M1)** as a baseline; the other two modes are collaborative modes based on co-embodiment, namely **Weighted-average-based virtual co-embodiment (M2)** and **Body-part-segmented virtual co-embodiment (M3)** (Figure 1).

Six participants participated in the study [Three males and three females, average age = 21.5 ± 3.5 (SD)]. They were informed about engaging in a collaborative task but were unaware of the specific form of collaboration, and they had either normal or corrected-to-normal vision. Pairings were established between a participant and one of the experimental members. Among the participants, four had no prior exposure to VR, one had limited prior experience, and one were already familiar with VR.

This study featured three modes in this cycling serious game. Firstly, in the instructional phase, the tutorial provided guidance for HMD usage and a demonstration of relevant operations. Subsequently, during the formal phase, participants randomly experienced three modes (M1, M2, and M3), performing game tasks with a 30-second rest between trials. In the post-task stage, participants removed HMDs for a 60-second break, responding to questionnaires. The 5-point Likert System Usability Scale (SUS) [5] was used to measure the usability of the designed serious game. The 5-point Likert Virtual Embodiment Questionnaire (VEQ) was used to measure the sense of agency (SoA) experienced by the participants [6]. Following all procedures, participants underwent interviews and provided demographic details. The entire process lasted about 50 minutes.

3 RESULTS AND DISCUSSION

3.1 System Usability

The results of the SUS are summarized in Table 1. The scores suggested that the usability of both collaborative modes (M2 and M3) in virtual co-embodiment was excellent (>85.0), acceptable (>70.0) and achieving A+ level (>84.1) [6]. Both collaborative modes scored higher than the Alone mode (M1), with the highest score (97.5) occurring in M2. It suggests that both collaborative modes in virtual co-embodiment exhibit high usability, potentially attributed to participants perceiving a heightened sense of engagement in these collaborative modes compared to the Alone mode, where the absence of collaborative feedback results in a slightly dull experience.

Table 1: System Usability Scale

Game modes	M1	M2	M3
Mean	84.58	85.83	87.50
Max.	95.00	97.50	95.00

3.2 SoA

Self-ratings Results of SoA The self-ratings of SoA are summarized in Table 2. The mean value indicates that the highest SoA rating occurs in Alone mode (M1) ($M = 5.08$, $SD = 1.87$), followed by the M2 ($M = 4.83$, $SD = 0.83$). Conversely, the lowest SoA rating is observed in the M3 ($M = 4.41$, $SD = 1.75$). A One-Way ANOVA analysis did not reveal significant differences in rating results across the three modes ($p = 0.666$). Therefore, the SoA in the two collaborative modes (M2 and M3) can reach a level close to that observed in the Alone mode.

Table 2: Self-ratings of SoA

Game modes	M1	M2	M3	p
Mean	5.08	4.83	4.41	0.666
SD	1.87	0.83	1.75	

Qualitative Results of SoA The SoA in the Alone mode (M1), when compared to two collaborative modes (M2 and M3), does not exhibit the anticipated significant difference. Based on the interviews, a possible explanation is that collaborative modes may lead participants to focus more intently on perceiving their control over the avatar. Hence, the perceived SoA might be slightly higher than under normal circumstances.

Based on interview results from participants on two collaborative modes, most of them consistently reported an elevated SoA when they perceived superior performance in a specific co-embodiment mode, despite the VEQ indicating comparable outcomes in both co-embodiment modes. There is a preliminary hypothesis that achieving a positive collaborative experience in co-embodiment modes, where partners are satisfied with each performance, may evoke an illusory SoA. The conjecture is that the collaborative motion modes with cooperation relationship in VR SGs might elicit a higher SoA among users to some extent. Besides, a preliminary inference from the feedback suggests that participants' prior VR experience may influence performance across modes, potentially impacting their perceived SoA in each mode to some extent.

4 CONCLUSION AND FUTURE WORK

This pilot study explored the perceived usability and agency of collaborative motion modes under virtual co-embodiment system in SGs. The results indicated that the new collaborative modes received a higher perceived usability. Considering quantitative and qualitative results, co-embodiment modes in this virtual serious game might induce a heightened user's SoA. Furthermore, qualitative feedback revealed potential enhancements to virtual co-embodiment based SGs design, prompting a plan for further investigation into user perceptions, such as attentional engagement and social presence. Possible areas for improvement in VR SGs with collaborative motion modes will be addressed through further diversiform data collection in future work.

REFERENCES

- [1] H. Liang and X. Dong, "Enhancing cognitive ability through a VR serious game training model mixing Piaget's epistemological methodology and Lumosity concept", *The Visual Computer*, Springer, vol. 38, no. 9-10, pp. 3487-3498, 2022.
- [2] A. Schena et al., "IAmHero: Preliminary Findings of an Experimental Study to Evaluate the Statistical Significance of an Intervention for ADHD Conducted through the Use of Serious Games in Virtual Reality," *International Journal of Environmental Research and Public Health*, MDPI, vol. 20, no. 4, 3414, 2023.
- [3] D. Kodama, T. Mizuho, Y. Hatada, T. Narumi, and M. Hirose, "Enhancing the Sense of Agency by Transitional Weight Control in Virtual Co-Embodiment," *International Symposium on Mixed and Augmented Reality (ISMAR)*, IEEE, pp. 278-286, 2022.
- [4] R. Fribourg et al., "Virtual Co-Embodiment: Evaluation of the Sense of Agency While Sharing the Control of a Virtual Body Among Two Individuals," *IEEE Transactions on Visualization and Computer Graphics*, IEEE, vol. 27, no. 10, pp. 4023-4038, 2021.
- [5] A. Bangor, P. T. Kortum, and J. T. Miller, "An Empirical Evaluation of the System Usability Scale," *International Journal Human-Computer Interaction*, Taylor & Francis Online, vol. 24, no. 6, pp. 574-594, 2008.
- [6] D. Roth and M. E. Latoschik, "Construction of the Virtual Embodiment Questionnaire (VEQ)," *IEEE Transactions on Visualization Computer Graphics*, IEEE, vol. 26, no. 12, pp. 3546-3556, 2020.